

UNIT V

SECURITY IN CLOUD COMPUTING

By
Dr. C P Ahir
LCO
GP Amravati

Cloud-Security consists of:-

user
provider

- ⇒ Set of Policies ✓
- ⇒ Controls ✓
- ⇒ procedures ✓
- ⇒ Technologies ✓

that work together to protect cloud
base systems, data & infrastructure.

⇒ Service delivery depends upon the
individual Cloud-Service Provider or the
Cloud Security Solutions.

Challenges/Issues:-

- 1) Data Loss ✓
- 2) Hacked interfaces and APIs
- 3- Data Breach (hacking of data)
- 4) Vendor lock-in
- 5) Denial of Service (DoS)
Attack (flooding of service
with so much traffic)
- 6) Account Hijacking ✓
- 7) Misconfiguration. }
- 8) Accidental Exposure of credentials
- 9) Cybersecurity Attacks
- 10) Data Privacy & Confidentiality

Security Risks of Cloud Computing

- * **Data Loss (Data Leakage) :** It is a the process in which data is being deleted, corrupted, and unreadable by a user, software, or application.
- * Data loss occurs when our sensitive data is somebody else's hands,
- * one or more data elements can not be utilized by the data owner,
- * hard disk is not working properly,
- * and software is not updated.

Security Risks of Cloud Computing

- * **Hacked Interfaces and Insecure APIs**
- * APIs are the easiest way to communicate with most of the cloud services.
- * In cloud computing, few services are available in the public domain.
- * These services can be accessed by third parties, so there may be a chance that these services easily harmed and hacked by hackers.

Security Risks of Cloud Computing

- * **Data Breach**

- * Data Breach is the process in which the confidential data is viewed, accessed, or stolen by the third party without any authorization, so organization's data is hacked by the hackers.

- * **Vendor lock-in**

- * the of the biggest security risks in cloud computing.
- * Organizations may face problems when transferring their services from one vendor to another.
- * As different vendors provide different platforms, that can cause difficulty moving one cloud to another.

Security Risks of Cloud Computing

- * **Increased complexity strains IT staff**
- * Migrating, integrating, and operating the cloud services is complex for the IT staff.
- * IT staff must require the extra capability and skills to manage, integrate, and maintain the data to the cloud.
- * **Spectre & Meltdown**
- * Spectre & Meltdown allows programs to view and steal data which is currently processed on computer.
- * It can run on personal computers, mobile devices, and in the cloud.
- * It can store the password, your personal information such as images, emails, and business documents in the memory of other running programs.

Security Risks of Cloud Computing

- * **Denial of Service (DoS) attacks**

- * Denial of service (DoS) attacks occur when the system receives too much traffic to buffer the server.
- * Mostly, DoS attackers target web servers of large organizations such as banking sectors, media companies, and government organizations.
- * To recover the lost data, DoS attackers charge a great deal of time and money to handle the data.

- * **Account hijacking**

- * It is the process in which individual user's or organization's cloud account (bank account, e-mail account, and social media account) is stolen by hackers. The hackers use the stolen account to perform unauthorized activities.

CLOUD RISK

- * Being an on-demand availability of system resources, like computing power and data storage, cloud computing involves various types of risks that are grouped in different categories like
 - * • **privacy , availability , changes and compliance**

CLOUD RISK

* 1. Privacy

- * • All of the below risks may result from malicious activities intended for attacking private data.
- * • **Controlled Access:** If the people/organization tries to store confidential data onto the cloud, the cloud's true nature provides access to the service provider organization.
- * • **Ownership Claim:** If the agreement is not well-read, the data's ownership can be unknowingly transferred to the service providing organization

CLOUD RISK

- * **2. Availability**

- * **• Service Disruption:** This can be attributed to any fault in the internet connection as all cloud computing transactions are done over the internet.
- * This can be either service quality degraded or outage as a whole.

CLOUD RISK

* 3. Changes

- * • **Change in Service:** Due to the volatile market, there may be acquisition or closure of a service provider, thus leading to unavailability of the service within short notice.
- * • **Return on investment:** The whole intuition behind cloud computing is to be cost-effective.
- * But due to unforeseen circumstances, the cost of the subscription is high;
- * it might jeopardize the whole purpose of cloud computing.

CLOUD RISK

* 4. Compliance

- * • **Audit:** The service provider organization might not follow the external audit process, thus leading to a vulnerable position for the end customers.
- * • **Storage Location:** Since the data for the services resides in hardware, and the location of that storage device is unknown, it might risk the country's sensitive data getting leaked by rival countries.
- * • **Notification:** Proper and transparent communication regarding lack of breach to the end customer puts them at risk as they might not be aware of the havoc caused due to the same.

CLOUD RISK DIVISON

- * Policy and Organizational Risks
- * Technical Risks
- * Legal risks

Policy and Organizational Risks

- * **Vendor Lock-in**
- * **Loss of Governance**
- * **Compliance challenges**
- * **Loss of business reputation due to co-tenant activities**
- * **Cloud services termination or failure**
- * **Cloud provider acquisition**
- * **Supply chain failure**

Policy and Organizational Risks

- * **Vendor Lock-in**

- * Organizations may face problems when transferring their services from one vendor to another.
- * As different vendors provide different platforms, that can cause difficulty moving one cloud to another.

- * **Loss of Governance**

- * The loss of governance in cloud computing occurs when businesses migrate workloads from an exclusively on-premises IT infrastructure to the cloud without a suitable governance policy in place.

Policy and Organizational Risks

* Loss of Governance

Performance Management

area of setting goals, responsibility accounting and monitoring/ analyzing/ governing and improving the performance

Risk management

process of identifying potential hazards to the business and acting to reduce or eliminate their financial impact.

Four Key elements

Value delivery

Value-Delivery involves everything necessary to ensure every paying customer is a happy customer: order processing, inventory management, delivery/fulfillment, troubleshooting, customer support, etc.

Strategic alignment

the process of planning and implementing practices to ensure an organization's strategies support its general objectives.

Policy and organizational risks

- * **Compliance challenges**
- * Several different industry regulations govern how organizations should manage and secure sensitive data.
- * Indian and foreign cloud service providers are subject to compliances under the Information Technology Act, 2000 ("**IT Act**") and corresponding rules on data privacy and law enforcement access.

Policy and organizational risks

- * **Loss of business reputation due to co-tenant activities**
- * **Resource sharing** means that malicious activities carried out by one tenant may affect the reputation of another tenant.
- * For example, spamming, port scanning or the serving of malicious content from cloud infrastructure can lead to:
 - * a range of IP addresses being blocked, including the attacker and other innocent tenants of an infrastructure;
 - * confiscation of resources due to neighbor activities

Policy and organizational risks

- * **Cloud Services termination or failure**
- * As in any new IT market, competitive pressure, an inadequate business strategy, lack of financial support, etc, could lead some providers to go out of business or at least to force them to restructure their service portfolio offering.
- * In other words, it is possible that in the short or medium term some cloud computing services could be terminated.
- * The impact of this threat for the cloud customer is easily understandable, since it could lead to a loss or deterioration of service delivery performance, and quality of service, as well as a loss of investment.

Policy and organizational risks

- * **Cloud Services termination or failure**
- * As in any new IT market, competitive pressure, an inadequate business strategy, lack of financial support, etc, could lead some providers to go out of business or at least to force them to restructure their service portfolio offering.
- * In other words, it is possible that in the short or medium term some cloud computing services could be terminated.
- * The impact of this threat for the cloud customer is easily understandable, since it could lead to a loss or deterioration of service delivery performance, and quality of service, as well as a loss of investment.

Policy and organizational risks

- * **Cloud Provider Acquisition**

- * Acquisition of the cloud provider could increase the likelihood of a strategic shift and may put non-binding agreements at risk (e.g., software interfaces, security investments, non-contractual security controls).

Policy and organizational risks

- * **Supply Chain Failure**

- * A cloud computing provider can outsource certain specialised tasks of its 'production' chain to third parties.
- * In such a situation the level of security of the cloud provider may depend on the level of security of each one of the links and the level of dependency of the cloud provider on the third party.
- * Any interruption or corruption in the chain or a lack of coordination of responsibilities between all the parties involved can lead to:
- * unavailability of services, loss of data confidentiality, integrity and availability, economic and reputational losses due to failure to meet customer demand, violation of SLA, cascading service failure, etc.

Technical Risk

- * **RESOURCE EXHAUSTION (UNDER OR OVER PROVISIONING)**
- * Inability to provide additional capacity to a customer: MEDIUM Comparative
- * Inability to provide current agreed capacity level
- * Service unavailability: failure in certain highly specific application scenarios which use a particular resource very intensively
- * Access control compromised: in some cases it may be possible to force a system to 'fail open' in the event of resource exhaustion
- * Economic and reputational losses: due to failure to meet customer demand.
- * Infrastructure oversize: excessive provisioning leading to economic losses and loss of profitability.

Technical Risk

- * **ISOLATION FAILURE**

- * Multi-tenancy and shared resources are two of the defining characteristics of cloud computing environments.
- * Computing capacity, storage, and network are shared between multiple users.
- * This class of risks includes the failure of mechanisms separating storage, memory, routing, and even reputation between different tenants of the shared infrastructure (e.g., so-called guest-hopping attacks, SQL injection attacks exposing multiple customers' data stored in the same table, and side channel attacks).

Technical Risk

- * **CLOUD PROVIDER MALICIOUS INSIDER - ABUSE OF HIGH PRIVILEGE ROLES**
- * The malicious activities of an insider could potentially have an impact on: the confidentiality, integrity and availability of all kind of data, IP, all kind of services and therefore indirectly on the organization's reputation, customer trust and the experiences of employees.

Technical Risk

- * **MANAGEMENT INTERFACE COMPROMISE (MANIPULATION, AVAILABILITY OF INFRASTRUCTURE)**
- * The customer management interfaces of public cloud providers are Internet accessible and mediate access to larger sets of resources (than traditional hosting providers)
- * and therefore pose an increased risk especially when combined with remote access and web browser vulnerabilities.

Technical Risk

- * **INTERCEPTING DATA IN TRANSIT**

- * Cloud computing, being a distributed architecture, implies more data in transit than traditional infrastructures.
- * For example, data must be transferred in order to synchronise multiple distributed machine images, images distributed across multiple physical machines, between cloud infrastructure and remote web clients, etc.
- * Furthermore, most use of data-centre hosting is implemented using a secure VPN-like connection environment, a practice not always followed in the cloud context.
- * Sniffing, spoofing, man-in-the-middle attacks, side channel and replay attacks should be considered as possible threat sources.

Technical Risk

- * **DATA LEAKAGE ON UP/DOWNLOAD, INTRA-CLOUD**

- * This is the same as the previous risk, but applies to the transfer of data between the cloud provider and the cloud customer.

- * **INSECURE OR INEFFECTIVE DELETION OF DATA**

- * Whenever a provider is changed, resources are scaled down, physical hardware is reallocated, etc, data may be available beyond the lifetime specified in the security policy.
- * It may be impossible to carry out the procedures specified by the security policy, since full data deletion is only possible by destroying a disk which also stores data from other clients.
- * When a request to delete a cloud resource is made, this may not result in true wiping of the data (as with most operating systems).
- * Where true data wiping is required, special procedures must be followed and this may not be supported by the standard API (or at all).

Technical Risk

- * **DISTRIBUTED DENIAL OF SERVICE (DDOS)**

- * There are several different scenarios in which a cloud customer's resources may be used by other parties in a malicious way that has an economic impact:
 - Identity theft: an attacker uses an account and uses the customer's resources for his own gain or in order to damage the customer economically.
 - The CC has not set effective limits on the use of paid resources and experiences unexpected loads on these resources through no malicious actions.
 - An attacker uses a public channel to use up the customer's metered resources - for example, where the customer pays per HTTP request, a DDoS attack can have this effect.

- **ECONOMIC DENIAL OF SERVICE (EDOS)**

- * EDoS destroys economic resources; the worst case scenario would be the bankruptcy of the customer or a serious economic impact.

Technical Risk

- * **LOSS OF ENCRYPTION KEYS**

- * This includes disclosure of secret keys (SSL, file encryption, customer private keys, etc) or passwords to malicious parties, the loss or corruption of those keys, or their unauthorised use for authentication and non-repudiation (digital signature).
- * **UNDERTAKING MALICIOUS PROBES OR SCANS**
- * Malicious probes or scanning, as well as network mapping, are indirect threats to the assets being considered.
- * They can be used to collect information in the context of a hacking attempt.
- * A possible impact could be a loss of confidentiality, integrity and availability of service and data.

Technical Risk

- * **COMPROMISE SERVICE ENGINE**

- * Each cloud architecture relies on a highly specialized platform, the service engine that sits above the physical hardware resources and manages customer resources at different levels of abstraction.
- * For example, in IaaS clouds this software component can be the hypervisor.
- * The service engine is developed and supported by cloud platform vendors and the open source community in some cases.
- * It can be further customized by the cloud computing providers.

Technical Risk

- * **CONFLICTS BETWEEN CUSTOMER HARDENING PROCEDURES AND CLOUD ENVIRONMENT**
- * Cloud providers must set out a clear segregation of responsibilities that articulates the minimum actions customers must undertake.
- * The failure of customers to properly secure their environments may pose a vulnerability to the cloud platform if the cloud provider has not taken the necessary steps to provide isolation.
- * Cloud providers should further articulate their isolation mechanisms and provide best practice guidelines to assist customers to secure their resources.
- * Customers must realize and assume their responsibility as failure to do so would place their data and resources at further risk.

Legal Risk

- * **SUBPOENA AND E-DISCOVERY**

- * In the event of the confiscation of physical hardware as a result of subpoena by law-enforcement agencies or civil suits , the centralisation of storage as well as shared tenancy of physical hardware means many more clients are at risk of the disclosure of their data to unwanted parties.

- * **RISK FROM CHANGES OF JURISDICTION**

- * Customer data may be held in multiple jurisdictions, some of which may be high risk. If data centres are located in high-risk countries
- * e.g., those. lacking the rule of law and having an unpredictable legal framework and enforcement, autocratic police states, states that do not respect international agreements, etc,

Legal Risk

- * **DATA PROTECTION RISKS**

- * Cloud computing poses several data protection risks for cloud customers and providers.
- * It can be difficult for the cloud customer to effectively check the data processing that the cloud provider carries out, and thus be sure that the data is handled in a lawful way.
- * There may be data security breaches which are not notified to the controller by the cloud provider.
- * The cloud customer may lose control of the data processed by the cloud provider. This issue is increased in the case of multiple transfers of data (e.g., between federated cloud providers).
- * The cloud provider may receive data that have not been lawfully collected by its customer (the controller).

Legal Risk

- * **LICENSING RISKS**

- * Licensing conditions, such as per-seat agreements, and online licensing checks may become unworkable in a cloud environment.
- * For example, if software is charged on a per instance basis every time a new machine is instantiated then the cloud customer's licensing costs may increase exponentially even though they are using the same number of machine instances for the same duration.

Legal Risk

- * **LICENSING RISKS**

- * Licensing conditions, such as per-seat agreements, and online licensing checks may become unworkable in a cloud environment.
- * For example, if software is charged on a per instance basis every time a new machine is instantiated then the cloud customer's licensing costs may increase exponentially even though they are using the same number of machine instances for the same duration.

Security concerns or Security issues in cloud computing

1. Confidentiality

6. Data Access

2. integrity

7. Data Segregation

3. Availability

8. Privacy

4. Authentication

9. Recovery

5. Authorization

10. Multi-tenancy

Data Security

1.Data Confidentiality

Data is only read by
intended recipient
(Passive attacks)

2.Data integrity

Data is not altered or
modified
(Active Attack)

3.Data Availability

Data available 24*4
(DOS attack)

What are the various technology used for data security in cloud computing?

1.Authentication and identity

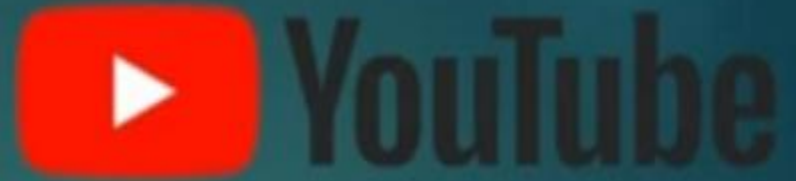
2.Authorization

3.Data Encryption

4.Data Masking

5.Information integrity and Privacy

6.Availability of information-SLA



Google



amazon

Google

Sign in with your Google Account



Email

Password

Sign in

☒ Stay signed in

[Need help?](#)



Home



Trending



Subscriptions



Library



History



Watch later



Liked videos



English speaking ...



Introduction to O...



Competitors

Related Movies



Raanjhanaa

From ₹25.00 CC



Ghanchakkar

From ₹50.00



Mary Kom

From ₹50.00 CC

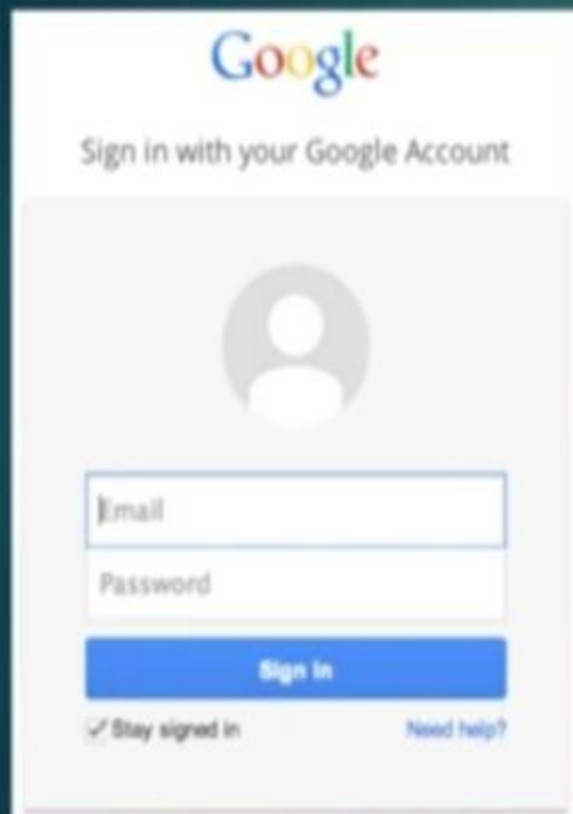


Madras Cafe

From ₹50.00 CC

importance of authentication in cloud computing

1. 2 Layer security



Google

Sign in with your Google Account



Email

Password

Sign In

☒ Stay signed in [Need help?](#)



facebook

Email or Phone Password Sign In

[Forgot account?](#)

Create an account

It's free and always will be.

First name Surname

Mobile number or email address

New password

Birthday

21 May 1994 Why do I need to provide my date of birth?

☐ Female ☐ Male

By clicking Sign Up, you agree to our Terms, Data Policy and Cookie Policy. You may receive SMS notifications from us and can opt out at any time.

Sign Up

importance of authentication in cloud computing

1. 2 Layer security

2. User id and Password

3. Validate the user

4. Protect from unauthenticated login attempts

5. Adds extra layer of security to cloud

importance of authorization in cloud computing

1.Access control

**2. Whether user is authorized for resources
and services**

3.Prevention from unauthorized access control

What is identity access management?

IAM is nothing but a framework of policies and technologies for ensuring that it providing the access of resources and services to the right people at the right time.

4 main componets of IAM

Authentication

Authorization

User Management

Central User
Repository

What is identity access management?

IAM is nothing but a framework of policies and technologies for ensuring that it providing the access of resources and services to the right people at the right time.

4 main componets of IAM

Authentication

- ✓ 2 layer security
- ✓ User id and pwd
- ✓ Session management

Authorization

- ✓ Access Control
- ✓ Allow or Deny access to a particular resources or services

User Management

- ✓ User credential management
- ✓ Session records of user

Central User Repository

- ✓ Data synchronization
- ✓ Store and manage IAM metadata related to particular user

What is Security as a Service (SECaaS)?

Cloud Service Models

A cloud service provider can provide infrastructure, platform, Software, etc., as a service.

For example, Google drive provides a file hosting service to a client. Here, Google Drive is an example of the Software as a Service cloud service model.



Security as a Service (SECaaS)

Security as a Service (SECaaS) is a business model in which a service provider provides security service to an individual or business on a subscription basis.

Why do we need Security as a Service (SECaaS)?

In the Security as a Service business model, a service provider often provides the security service in a cost-effective manner.

A client does not need any specific on-premises hardware. Instead, the client gets the security service on a subscription basis.

Services provided by the Security as a Service (SECaaS) Business Model

Different types of security services can be provided by the service provider.

For example, a service provider can provide anti-malware programs, firewalls, Intrusion Detection Systems (IDS), authentication services, or even penetration testing services.

A service provider can also provide business continuity and disaster recovery service, Data Loss Prevention (DLP) service, email security, Identity, and Access Management service, vulnerability scanning service, etc.



Security Services

➤ X.800 divides the security services into five categories.

- Authentication
- Access control
- Data confidentiality
- Data integrity
- Non-repudiation

